
DECOMPRESSING A STORY OF CLUMPING

The Grammar of Clumping

Five cartoons that choose a scale, and what the exponents leave behind

Every scaling law a physicist teaches is a cartoon: a power law fitted to a real threshold, standing in for a mechanism too intricate to draw in full. The cartoons are not thereby worthless. They are the residue that remains once the derivation is forgotten, and residues are exactly what this essay is trying to take seriously.

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Decompressing *The Making of Goldilocks: A Story of Clumping* (2020)

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Full citations, checked numbers, and a reading route included

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The argument in one paragraph

Matter clumps for at least three unrelated reasons, and each reason spends a different currency of binding energy at a different order of magnitude, which is why the clumped objects it produces come out at such different scales. Gravity, the weakest force by a margin of roughly thirty-six orders of magnitude, wins only once enough atoms are gathered that its whispered pull, summed over all of them, outshouts the shout of electrostatic bonding; that threshold sits somewhere in the range of large asteroids to small moons, a few hundred kilometres in radius, though a naive energetic argument for it, of the kind this essay starts with, overshoots the true figure for exactly the reason a much more careful argument by Lineweaver and Norman had to correct in itself. A living clump spends a second currency, chemical energy shipped through a fractal network of vessels, and the exponent this yields, three-quarters rather than two-thirds, has been argued over for ninety years by half a dozen incompatible mechanisms that all reproduce it, which is itself evidence that none of them has yet been shown hard to vary in Deutsch’s sense. A third currency, thermal pressure resisting gravity’s pull on a whole cloud, sets the Jeans mass that once separated the primordial plasma into precursors of galaxies, an argument classical enough to misapply, and grand enough to misattribute, since it belongs to Jeans and not, as habit sometimes has it, to Rayleigh, whose own criterion governs a different kind of rotating instability entirely. A fourth threshold is not about binding energy at all but about statistical reliability: Schrödinger’s observation that organisms must be enormous compared to their atoms so that the law of large numbers can do the disciplining that individual atoms, left to their thermal jitter, cannot do for themselves. What survives, once the exponents themselves are forgotten, is the shape common to all four: a currency, a threshold, and a scale that falls out of the two by nothing more than dimensional bookkeeping.

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1 A residue, and the cartoons it is made of

Ask a physicist why the Earth is round and a good answer arrives in a sentence: gather enough atoms and gravity’s cumulative pull eventually beats the electrostatic bonds that give a small rock its lumpy, arbitrary shape. Ask a physiologist why an elephant’s metabolism does not simply scale with its skin, and a good answer arrives almost as quickly: a fractal network of vessels delivers fuel more efficiently than a flat surface ever could. Both answers are cartoons, in the specific sense this essay wants to defend rather than apologise for: a cartoon compresses a real mechanism into a shape simple enough to reason with, at the cost of details that would only clutter the point being made. The test of a good cartoon is not whether it omits detail, since every explanation does that, but whether the details it keeps are the ones doing the work, which is close to Deutsch’s demand that a good explanation be hard to vary [Deutsch, 2011].

This essay decompresses a short piece written in 2020, *The Making of Goldilocks: A Story of Clumping*, which compressed three such cartoons, electrostatic-versus-gravitational binding, Kleiber’s metabolic exponent, and the Jeans mass of a collapsing cloud, into a few paragraphs and a handful of equations [McCulloch-James, 2020]. Decompression here means three things: working the derivations through in full rather than in shorthand, checking the numbers that the original left as round figures, and being honest about where a cartoon’s own arithmetic runs ahead of what it can support. A fourth cartoon, Schrödinger’s argument for why organisms are the size they are, did not appear in the original piece and is added here because it clumps by an entirely different currency, statistical reliability rather than binding energy, and the contrast is instructive. Boxes marked *a useful compression* give the original’s own words or working; boxes marked *decompression handle* do the expanding.

2 The grammar of a scaling argument

Every argument in this essay has the same skeleton, and it is worth exposing the skeleton once so that four different bodies do not obscure the single joint they share.

1. **Fix a shape.** Assume constant density ρ and a roughly spherical body, so that mass and radius are tied together: $M \propto r^3$, hence $r \propto M^{1/3}$.
2. **Name a currency.** Identify the energy, force, or statistical quantity that is actually doing the work of holding the clump together or driving it apart.
3. **Write the currency as a function of M .** Substituting $r \propto M^{1/3}$ into the currency’s own formula turns a statement about radius into a statement about mass, and produces a power law.
4. **Compare two currencies.** A clump forms, persists, or changes character at the mass where one currency’s power law overtakes another’s, and since the two typically carry different exponents, the crossover happens at one particular scale rather than at all scales or none.

That skeleton is dimensional analysis with a narrative attached, and its virtue is exactly the virtue of a cartoon: it forces an exponent out of geometry alone, before any detailed physics has been supplied, which is why the same four steps recur, with a different currency substituted each time, across gravity, metabolism, star formation, and heredity.

3 First clump: when gravity outguns the electron

A useful compression

The original argument runs: for constant density, $V \propto r^3$ and $M \propto r^3$, so $M^{1/3} \propto r$. The work done to remove an atom from a body's own self-gravity is $W = Fr = Mr^{-2} \cdot r = Mr^{-1}$, so $W \propto r^{-1} \propto M^{-1/3}$. Planets composed of the order of 10^{54} atoms have a mass sufficient to compensate for gravity's 10^{36} orders of weakness relative to the electrostatic force [McCulloch-James, 2020].

3.1 Decompressing the derivation

The quantity W in the original is a per-particle energy, not the total binding energy of the body, and making that explicit removes the appearance of a dropped constant. Pull one atom of mass m away from a spherical body of mass M and radius r ; the work done against the body's own gravity is

$$W(M) = \frac{GMm}{r} = Gm \left(\frac{4\pi\rho}{3} \right)^{1/3} M^{2/3},$$

using $r = (3M/4\pi\rho)^{1/3}$ for constant density ρ . The exponent $2/3$ falls out of the geometry alone, exactly as the original claims; what the compression left implicit is that m , ρ , and G are being held fixed while M varies, and that W is an energy *per atom removed*, to be compared against the roughly constant per-atom binding energy of ordinary chemical bonds, typically a few electron-volts. A body clumps into a sphere once gravity's per-atom bill exceeds that fixed chemical price; below the threshold, an irregular, electrostatically rigid shape survives collision and accretion without penalty.

Decompression handle

Two numbers in the original are worth putting through a calculator rather than leaving as round figures. The ratio of the electrostatic to the gravitational force between two protons is

$$\frac{e^2/4\pi\epsilon_0}{Gm_p^2} \approx 1.24 \times 10^{36},$$

which confirms the quoted 10^{36} almost exactly. The atom count is less forgiving. Earth's actual composition, roughly iron, oxygen, silicon, and magnesium with a mean atomic mass near 30, gives an atom count of

$$N_{\oplus} = \frac{M_{\oplus}}{\bar{m}} \approx \frac{5.97 \times 10^{24} \text{ kg}}{5.0 \times 10^{-26} \text{ kg}} \approx 1.3 \times 10^{50},$$

four orders of magnitude short of the quoted 10^{54} . The figure of 10^{54} is recovered instead if the reference particle is hydrogen rather than rock: 10^{54} hydrogen atoms weigh $1.67 \times 10^{27} \text{ kg}$, which is 0.88 Jupiter masses, a gas giant rather than a rocky planet. The original sentence is not wrong so much as underspecified: it names the right order of magnitude for *some* planetary body while leaving open which body and which atom, and the two choices differ by four decades. A cartoon can survive that kind of looseness; a number quoted to four significant figures could not.

3.2 Where the per-atom argument overshoots

Pushing the per-atom energy criterion to its crossover mass, using a rocky density of 3000 kg m^{-3} and a chemical bond strength of a few electron-volts, gives a crossover radius in the range of two to

five thousand kilometres, roughly a fifth of an Earth mass. That is the right order of magnitude for a planetary body and the wrong order of magnitude for the actual transition. Planetary scientists calibrate the true threshold, sometimes called the potato radius, from the observed shapes of asteroids and icy moons rather than from a bare energy comparison, and they find it far smaller: about 300 km for rocky bodies and 200 km for icy ones, based on the material's yield strength rather than its raw binding energy [Lineweaver and Norman, 2010].

Provenance check

Lineweaver has recounted the false start his own calculation made before landing on this figure, and it is worth keeping in a classroom precisely because it is a professional's mistake rather than a beginner's. His first calculation asked what surface gravity would be needed to deform rock at a planet's surface, and returned an answer of roughly ten thousand kilometres, which is absurd on its face, since bodies far smaller than that are observably round. The error was not in the arithmetic but in the question: rock is not crushed at the surface, where the original calculation was looking, but at depth, under the overburden pressure of everything above it, and asking about overburden pressure rather than surface gravity brought the figure down to the observed few hundred kilometres. The per-atom energy comparison in this section makes a milder version of the identical mistake, comparing a bulk average to a local threshold, which is why it overshoots in the same direction for the same underlying reason.

The lesson generalises past planets. A dimensional argument built from the right variables gets the exponent right by construction, since the exponent is forced by the geometry; it can still get the crossover *value* wrong by a wide margin if the physical quantity being compared is measured at the wrong location or averaged when it should have been evaluated locally. Exponent and prefactor are not the same kind of claim, and a cartoon is far more trustworthy about the first than the second.

4 Second clump: Kleiber's exponent and the crowd of rivals

A useful compression

The original notes that a live body, modelled as a sphere radiating heat through its surface area $A = 4\pi r^2$, would be expected to metabolise at a rate proportional to $M^{2/3}$. The basal metabolic rate instead follows Kleiber's law, $U \propto M^{3/4}$, which the piece attributes to modelling the body as a higher-dimensional sphere with fractal-like inner structure [McCulloch-James, 2020].

4.1 Decompressing Rubner's surface law

Max Rubner demonstrated in 1883, in respiration trials on dogs, that basal heat production scales with body surface area rather than body mass, an entirely sensible starting hypothesis for a warm-blooded animal shedding heat through its skin [Rubner, 1883]. Since $A \propto r^2$ and $r \propto M^{1/3}$ for constant density, this predicts $U \propto M^{2/3}$, the exponent the compressed passage above starts from. For decades this surface law was the textbook answer.

4.2 Decompressing Kleiber's correction, and the crowd it provoked

Max Kleiber, an agricultural chemist working on animal metabolism at Davis, plotted the basal metabolic rates of mammals from rats to steers against their body mass on logarithmic axes in 1932 and found a slope close to three-quarters rather than two-thirds [Kleiber, 1932]. The empirical exponent has held up across an extraordinary range of organisms since, from microbes to whales, and Savage et al. [2004] document its predominance across biology, yet the reason for three-quarters

rather than two-thirds remains contested in a way the reason for the electrostatic-versus-gravity threshold in the previous section does not.

Decompression handle

West, Brown, and Enquist proposed that the three-quarter exponent emerges from the geometry of a space-filling fractal network of branching vessels, xylem in a plant, capillaries in a mammal, optimised so that the energy dissipated in transporting nutrients is minimised [West et al., 1997]; a fourth spatial dimension, effectively, is what the fractal branching buys the organism. This is close to the original piece’s own gesture at “hyper-spheres” and “fractal-like inners,” and it is the account most often taught. It is not, however, the only one on offer, nor is it uncontested even on its own mathematical terms. Kozłowski and Konarzewski have argued that the West-Brown-Enquist derivation contains assumptions that do not actually force the three-quarter exponent, and that the model is less biologically constrained than it appears [Kozłowski and Konarzewski, 2004]. McMahon’s earlier and independent proposal, elastic similarity in the scaling of limbs against the risk of buckling, arrives at the same exponent by a route that shares no obvious mechanism with fractal vessels. And White and Seymour, re-examining basal metabolic rate data with corrections for measurement conditions, argued that the true mammalian exponent is close to two-thirds after all, reopening a question many had thought closed [White and Seymour, 2003].

Provenance check

This is the moment in the essay to apply Deutsch’s test rather than merely cite it. A good explanation is one that would have failed had the world been different, its components load-bearing enough that swapping one out breaks the fit to data [Deutsch, 2011]. Kleiber’s exponent has instead attracted several structurally unrelated derivations, fractal transport networks, elastic buckling limits, and a contested re-measurement that questions whether the number itself is even three-quarters, all converging on roughly the same empirical slope. Convergence of that kind is not evidence that biology has found an especially robust law; it is closer to the diagnostic Deutsch himself warns about, since an explanation that several incompatible mechanisms could equally have produced is, to that same degree, easy to vary. The exponent survives; the story behind it, unlike the electrostatic threshold’s, is still genuinely up for grabs.

D’Arcy Thompson’s older observation belongs in this company as well: growth and form are constrained by the same physical scaling that constrains a cathedral or a skeleton, so that no organism is merely a scaled photograph of another [Thompson, 1917]. Kleiber’s exponent is one instance of a much older insistence, that shape has consequences, without yet being a settled instance of why this particular consequence takes the value it does.

5 Provenance check: Jeans, not Rayleigh

Before decompressing the third clump, a correction is owed to the story itself, since the criterion that governs when a self-gravitating cloud collapses is due to James Jeans, not Lord Rayleigh, and the two names belong to different problems that happen to share an era and a habit of stability analysis.

Provenance check

The gravitational collapse criterion used below is Jeans’s, from his 1902 paper on the stability of a spherical nebula, which asks whether a cloud’s internal thermal pressure can resist the pull of its own gravity [Jeans, 1902]. Rayleigh’s own contribution to rotating astrophysical

fluids is a different criterion entirely, published in 1917, which asks whether a differentially rotating fluid is stable against axisymmetric interchange, and answers that it is stable exactly when specific angular momentum increases outward, $d(r^2\Omega)^2/dr > 0$ [John William Strutt, 1917]. Jeans's criterion involves no rotation at all; Rayleigh's involves no self-gravity as its subject; the confusion is an easy one to make, since both are named stability criteria from Cambridge-trained physicists of the same generation, applied to spinning or self-gravitating fluids, but they are not interchangeable and answer different questions. Rayleigh's own better-known contribution to the colour of the sky, discussed elsewhere in this project, makes the slip even more understandable, since it puts his name beside skies and stars in the same imaginative neighbourhood that Jeans's mass occupies.

6 Third clump: the Jeans mass at the edge of an expanding universe

A useful compression

Considering a mass accretion of radius r and mass M , the gravitational potential energy is $E_p = -GM^2/r$ and the internal energy is $U \propto r^3 p$. Gravitational clumping is favoured when $E_p + U \approx 0$, giving a critical Jeans mass $M_J \propto \sqrt{p^3/\rho^4}$. At recombination, with $T = 3000$ K and $\rho \approx 9.9 \times 10^{-19} \text{ kg m}^{-3}$, modelling the relativistic plasma pressure as $p_R \approx \frac{1}{3}c^2\rho$, the resulting Jeans mass is $9.7 \times 10^{48} \text{ kg}$, about five billion billion solar masses [McCulloch-James, 2020].

6.1 Decompressing the balance

Jeans's own criterion asks when a perturbation in a self-gravitating gas grows rather than oscillates, and the energetic shorthand above captures its spirit: a cloud collapses once its gravitational potential energy overwhelms the thermal pressure resisting compression [Jeans, 1902]. Carrying the algebra through for constant density, using $M \propto r^3\rho$ to eliminate r in favour of M , produces

$$M_J = \frac{1}{G^{3/2}} \frac{p^{3/2}}{\rho^2},$$

the formula the original quotes. Recomputing the recombination-era estimate independently, using the same inputs, returns $M_J \approx 9.6 \times 10^{48} \text{ kg}$, or 4.8×10^{18} solar masses, matching the original's figure closely enough that the arithmetic, at least, checks out.

Provenance check

The derivation is nonetheless a cartoon nested inside a cartoon, and it is worth saying so plainly. Jeans's own criterion is Newtonian, static, and non-rotating, derived for an isothermal gas cloud sitting still in otherwise empty space. Applying it to the primordial plasma at recombination, which is relativistic, expanding, and coupled to radiation rather than isothermal, borrows the shape of the argument while working well outside the conditions under which it was derived. Modern treatments of structure formation use a Jeans length adapted to an expanding, radiation-dominated background and full transfer functions rather than this single algebraic balance. None of that makes the exercise worthless: a cartoon applied outside its warranty can still return the right order of magnitude, and the fact that a classical instability criterion, invented to describe interstellar nebulae, gives a roughly sensible mass scale for cosmic structure eighty years before the physics of recombination was understood in detail is itself a small argument for keeping cartoons around. It is not, however, a substitute for the

machinery that has since replaced it, and a class should be told which job the simple formula is and is not doing.

6.2 The DNA analogy, held loosely

The original piece closes this thread with an evocative comparison: humans are composed of order 10^{17} DNA molecules, which it suggests is of the same order as the ratio between the Jeans mass at recombination and the mass of the Sun. That ratio is close to 5×10^{18} , roughly forty times the quoted DNA figure, so the comparison holds only to within a couple of orders of magnitude rather than as an exact match. The honest way to keep the analogy is as a gesture rather than a result: a human is to their genetic material roughly as the Jeans mass of the infant universe was to a single star, a vast structure organised, in each case, around a comparatively minuscule store of instructions. The parallel is worth remembering for its shape and not for its digits.

7 A different currency: Schrödinger's order from disorder

The first three clumps are all energy-balance arguments: a currency of binding energy is compared against a rival, and a mass falls out where the two are equal. Schrödinger's argument for the size of organisms belongs to a different family entirely, since its currency is not energy but statistical reliability, and it is worth adding here precisely because the contrast sharpens what the earlier three have in common.

Decompression handle

In *What Is Life?*, delivered as lectures in Dublin in 1943 and published the following year, Schrödinger asks why organisms, and the sense organs and machinery inside them, are so enormously large compared to the atoms they are built from [Schrödinger, 1944]. His answer rests on the $\sqrt{N}$ law: physical quantities built from the collective behaviour of N independent particles fluctuate relative to their mean by roughly $1/\sqrt{N}$, so that a system needs a very large N before its behaviour becomes reliably lawlike rather than dominated by thermal noise. An organ, a nerve, or a sensory receptor built from only a few thousand molecules would be at the mercy of exactly the kind of statistical jitter that a mole of gas comfortably averages away; reliable physiology therefore requires organisms, and their functional parts, to contain enormous numbers of atoms; Schrödinger calls this principle order-from-disorder, since a macroscopically dependable order emerges from microscopically undisciplined chaos purely through the arithmetic of large numbers.

He then poses the puzzle this creates for heredity. Using Delbrück's radiation experiments, which estimated that a single gene comprises only a few thousand atoms, Schrödinger notes that a structure this small ought, on the $\sqrt{N}$ argument alone, to be hopelessly unstable against thermal fluctuation, and yet inherited traits persist with striking fidelity across generations. His proposed resolution invokes a second principle, order-from-order: chemical bonds are quantum mechanically stable in a way that statistical averaging is not, so that a small but chemically bonded structure can be as reliable as a much larger, merely statistical one. He named the hypothetical carrier of heredity an aperiodic crystal, on the reasoning that a periodic crystal, repeating the same small unit indefinitely, could not encode enough distinct information to specify an organism, while an aperiodic one, varying along its length, could.

Provenance check

Schrödinger’s specific mechanism for gene stability, quantum-mechanical bond rigidity alone, was not quite the answer that emerged a decade later. The double helix, when Watson and Crick described it, achieves its fidelity through complementary base pairing and enzymatic proofreading during replication, a chemical and informational architecture considerably richer than “a stable molecule” by itself. Schrödinger’s aperiodic-crystal guess about the *shape* heredity would need to take was strikingly prescient; his account of *why* that shape would be stable was less so. The $\sqrt{N}$ argument for organism size, by contrast, needed no revision at all, and stands today essentially as he stated it. A single book can therefore contain one argument that aged perfectly and one that needed replacing, which is a useful thing for a student to sit with before assuming that being right about the big picture means being right about the mechanism.

8 Three currencies of binding

The three energy-balance clumps can now be placed side by side, since each spends a different currency, and the currencies themselves differ by enormous factors that explain, without further argument, why planets, organisms, and stars sit at such different scales.

Currency	Typical scale	What it binds	Threshold it sets
Gravitational (self, per atom)	fractions of an eV to tens of eV, depending on body size	Any mass, however diffuse, given enough of it	Potato radius: a few hundred km, where self-gravity exceeds material yield strength [Lineweaver and Norman, 2010]
Chemical / electrostatic	a few eV per bond	Molecules, crystalline solids, small bodies before self-gravity dominates	The very threshold gravitational binding must exceed to round a body, or metabolism must ship to sustain a cell [West et al., 1997]
Nuclear (fusion)	millions of eV per nucleon	Atomic nuclei; ignites only once gravitational collapse has raised core temperature and density far enough	Converts a Jeans-unstable, still-collapsing cloud into a star in hydrostatic equilibrium, the collapse halted by radiation pressure from fusion rather than by ordinary gas pressure alone

The nuclear row is the hinge connecting the first clump to the third. A cloud that crosses the Jeans threshold does not become a star merely by collapsing; gravitational collapse converts potential energy into heat, raising the core’s temperature and density, and only once that core reaches a few million kelvin does the nuclear currency, roughly six orders of magnitude richer per particle than the chemical currency, switch on and supply the pressure that halts further collapse. A body that collapses under gravity but never reaches ignition, a brown dwarf or a gas giant, is arrested instead by electron degeneracy pressure, a quantum mechanical effect with no analogue in either the chemical or the classical gravitational story told above. The three currencies, and the quantum coda to them, are what separate a planet from a star from the inert rubble in between, and the separation is a matter of which binding energy gets spent first, not of any difference in the raw material.

9 What the cartoons have in common

Set side by side, the four arguments share a family resemblance worth naming explicitly. Each fixes a shape, names a currency, derives a power law from geometry alone, and finds a crossover mass where two currencies balance. Each compresses a great deal of mechanism into a single exponent, and each exponent is more trustworthy than the prefactor multiplying it, since the exponent follows from dimensional bookkeeping that no detail of the mechanism can override, while the prefactor depends on exactly the local details, overburden pressure rather than surface gravity, an expanding relativistic background rather than a static isothermal one, that a first pass at the argument is most likely to get wrong.

The electrostatic-versus-gravity threshold and the Jeans mass are, in Deutsch’s sense, comparatively hard to vary: swap out the physics and the prediction breaks in a way a real measurement could catch, which is why both survive as the standard first explanations they are, adjusted at the edges rather than abandoned [Deutsch, 2011]. Kleiber’s exponent occupies a more uncomfortable position, a number robust across a century of measurement while its explanation remains contested among several structurally unrelated mechanisms that fit equally well, which is closer to the diagnostic of an explanation too accommodating rather than too constrained. And Schrödinger’s $\sqrt{N}$ argument is neither an energy balance nor, on its own admission, the whole story, since it explains why organisms must be large without yet explaining, by itself, how a large organism assembles the small, reliable parts, genes, that a purely statistical argument would call impossible.

10 Coda: the residue after the exponents are forgotten

A student who forgets every exponent in this essay, the two-thirds, the three-quarters, the messier power inside the Jeans mass, and the one-half buried in $\sqrt{N}$, has not necessarily forgotten anything that mattered. What is worth keeping is the shape underneath all four: that a clump is not a fact about matter in general but a settlement between two rival currencies, that a currency has a natural per-particle scale, and that the size of the settlement is forced, almost before any detailed physics is supplied, by nothing more than the geometry connecting mass to radius. A cartoon of that kind survives being forgotten in its particulars precisely because its particulars were never the point; the point was always the grammar, gravity against electrostatics, transport against surface, pressure against self-gravity, statistics against thermal noise, and the grammar is what a good cartoon is built to leave behind.

None of this licenses treating every power law as equally trustworthy. Lineweaver’s own false start, the ten-thousand-kilometre answer to the wrong question about surface gravity, is a professional’s demonstration that a correct derivation from the wrong variable produces a confident, well-formed, entirely mistaken number, and Kleiber’s contested exponent is a demonstration that an empirically secure number can outlast every attempt so far to explain why it takes the value it does. The residue that survives forgetting is not a licence to stop checking; it is what checking, done often enough, eventually leaves standing.

11 A reading route for students and teachers

1. **Start with the original.** Read McCulloch-James [2020] in full before this decompression, and mark every place a round number appears without its working shown.
2. **Rebuild the electrostatic threshold.** Derive $W(M) \propto M^{2/3}$ from scratch, then compute the crossover mass for a chemical bond strength of your own choosing, comparing your answer against the potato radius of Lineweaver and Norman [2010].

3. **Follow the Kleiber dispute.** Read Kleiber [1932] for the original data, West et al. [1997] for the fractal-network explanation, and White and Seymour [2003] for the case that the exponent itself is two-thirds after all. Decide, for yourself, which explanation is easiest to vary.
4. **Untangle Jeans from Rayleigh.** Read the abstract of Jeans [1902] and, alongside it, any modern derivation of Rayleigh’s rotating-fluid criterion, and write one sentence stating what each criterion assumes that the other does not.
5. **Read Schrödinger on his own terms.** Read the chapter “Why Are the Atoms So Small?” in Schrödinger [1944], and separate, in your own words, the argument that has aged well from the one that has not.
6. **Compare the currencies.** Using Rees’s discussion of the constants that fix stellar and planetary scales, work out roughly what core temperature is needed to ignite hydrogen fusion, and place that alongside the chemical and gravitational thresholds discussed above [Rees, 2000].
7. **Read the cosmological context.** Weinberg’s account of the early universe gives the standard picture of the epochs this essay’s Jeans-mass calculation moves through in a single line [Weinberg, 1977].

Question left open

Every threshold in this essay is a crossover between exactly two currencies. Real clumps rarely restrict themselves to two: a forming star is shaped by gravity, gas pressure, radiation pressure, magnetic fields, and rotation all at once, and a living organism by chemistry, statistics, and mechanical constraint together. What is lost, and what is gained, by teaching the two-currency cartoon first?

References

- David Deutsch. *The Beginning of Infinity: Explanations That Transform the World*. Allen Lane, London, 2011. ISBN 9780713992748.
- James Hopwood Jeans. The stability of a spherical nebula. *Philosophical Transactions of the Royal Society of London A*, 199:1–53, 1902. doi:[10.1098/rsta.1902.0012](https://doi.org/10.1098/rsta.1902.0012).
- Lord Rayleigh John William Strutt. On the dynamics of revolving fluids. *Proceedings of the Royal Society of London A*, 93(648):148–154, 1917. doi:[10.1098/rspa.1917.0010](https://doi.org/10.1098/rspa.1917.0010).
- Max Kleiber. Body size and metabolism. *Hilgardia*, 6(11):315–353, 1932. doi:[10.3733/hilg.v06n11p315](https://doi.org/10.3733/hilg.v06n11p315).
- Jan Kozłowski and Marek Konarzewski. Is west, brown and enquist’s model of allometric scaling mathematically correct and biologically relevant? *Functional Ecology*, 18(2):283–289, 2004. doi:[10.1111/j.0269-8463.2004.00830.x](https://doi.org/10.1111/j.0269-8463.2004.00830.x).
- Charles H. Lineweaver and Marc Norman. The potato radius: A lower minimum size for dwarf planets. *Proceedings of the 9th Australian Space Science Conference*, pages 67–78, 2010. URL <https://arxiv.org/abs/1004.1091>.
- Lee McCulloch-James. The making of goldilocks: A story of clumping. *McMurmerings*, 2020. URL <https://mcmurmerings.wordpress.com/2020/02/21/the-making-of-goldilocks-a-story-of-clumping/>. Published 21 February 2020; accessed 18 September 2026.
- Martin Rees. *Just Six Numbers: The Deep Forces That Shape the Universe*. Basic Books, New York, 2000. ISBN 9780465036722.
- Max Rubner. Ueber den einfluss der körpergrösse auf stoff- und kraftwechsel. *Zeitschrift für Biologie*, 19:535–562, 1883.
- Van M. Savage, James F. Gillooly, William H. Woodruff, Geoffrey B. West, Andrew P. Allen, Brian J. Enquist, and James H. Brown. The predominance of quarter-power scaling in biology. *Functional Ecology*, 18(2):257–282, 2004. doi:[10.1111/j.0269-8463.2004.00856.x](https://doi.org/10.1111/j.0269-8463.2004.00856.x).
- Erwin Schrödinger. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press, Cambridge, 1944.
- D’Arcy Wentworth Thompson. *On Growth and Form*. Cambridge University Press, Cambridge, 1917.
- Steven Weinberg. *The First Three Minutes: A Modern View of the Origin of the Universe*. Basic Books, New York, 1977. ISBN 9780465024377.
- Geoffrey B. West, James H. Brown, and Brian J. Enquist. A general model for the origin of allometric scaling laws in biology. *Science*, 276(5309):122–126, 1997. doi:[10.1126/science.276.5309.122](https://doi.org/10.1126/science.276.5309.122).
- Craig R. White and Roger S. Seymour. Mammalian basal metabolic rate is proportional to body mass^{2/3}. *Proceedings of the National Academy of Sciences of the United States of America*, 100(7):4046–4049, 2003. doi:[10.1073/pnas.0436428100](https://doi.org/10.1073/pnas.0436428100).